Today’s agenda

- Assignment07 presentation
- FreeRTOS recap
- Queues recap
- Deadlocks / queue topology
- Introduction to Assigment08
- Lab09

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Course outline

- **The plan is indicative, subject to change!**

Date	Lecture	Subject(s)	Lab	Assignment due
Feb 2	1	Introduction; ARM Cortex-M4	Lab1: Setting up the development env.	
Feb 9	2	EMP coding standard; Bit manipulation	Lab2: Bit manipulation	
Feb 16	3	State machines; The compiler chain; EMP-Board	Lab3: State machines	PF1 – Assignment 1
Feb 23	4	The task model; The pre-processor	Lab4: A clock radio task model	PF1 – Assignment 2
Mar 2	5	Queues and semaphores; Debugging	Lab5: Debug with a serial connection	PF1 – Assignment 3
Mar 9	6	Run to complete scheduler	Lab6: RTCS, Run to compile scheduler	PF1 – Assignment 4
Mar 16	7	More debugging; C: printf()	Lab7: RTCS, Debugger	PF1 – Assignment 5
March 23	8	FreeRTOS	Lab8: FreeRTOS	PF1 – Assignment 6
March 30 April 6		Easter holiday		
Apr 13	9	Re-entrance, more queues	Lab9: FreeRTOS (continued)	PF1 – Assignment 7
Apr 20	10	Assembler in C,	Work on the final assignment	PF1 – Assignment 8
Apr 27	11	Work on the final assignment, consultations		
May 4	12	Work on the final assignment, consultations		
May 11	13	Poster session		PF2- Final Assignment

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Notes about final assignment

- Work needs to be done WITHIN groups
- Work cannot be done between groups
- Code cannot be taken from any other student groups/years
- If code is taken from some online solution, this needs to be discussed with teacher beforehand and mentioned in the report
- Absolutely no ChatGPT, Claude, or other generative AI solutions
- If any of the above points are not adhered to, it will be treated as plagiarism

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Recap - FreeRTOS

- RTOS
  - Real-time operating system
  - Responds to events within a strictly defined time (deadline)
  - Processes data as it comes in without delay
  - Scheduler is deterministic
  - Used when
    - Meeting deadlines is critical for task completion
    - Results only make sense if delivered within time constraints

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Recap - FreeRTOS

- FreeRTOS
  - Small-enough for microcontrollers
  - Provides only
    - Real-time scheduling
    - Inter-task communication
    - Timing and synchronization primitives
  - More a real-time kernel than OS

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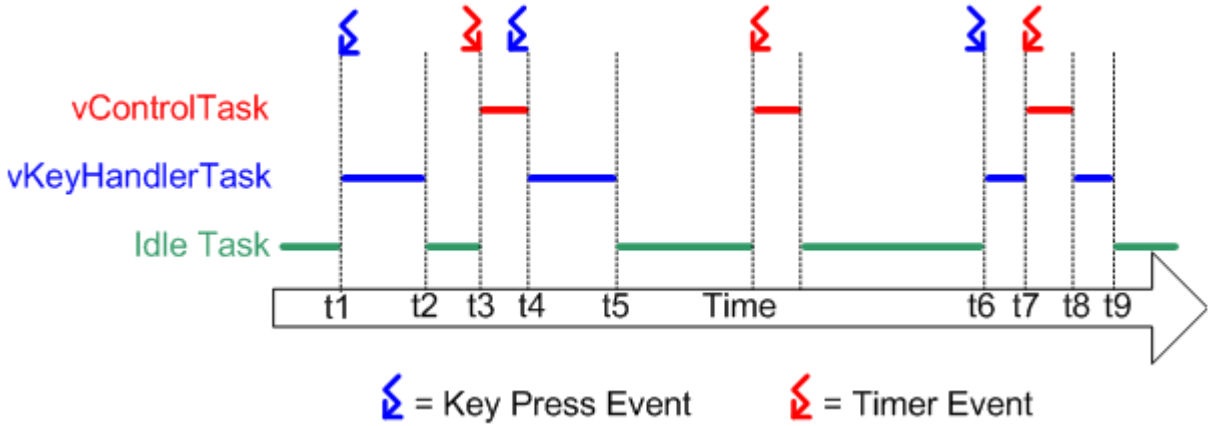
Recap - FreeRTOS



- Scheduler
  - Part of kernel deciding on task execution
  - Multi-tasking – executing tasks seemingly simultaneously
  - Pre-emptive scheduling
    - Runs tasks with highest priority first
    - Tasks of identical priority share CPU time (round robin time slicing)
  - Task states: running, ready, blocked, suspended

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Recap - FreeRTOS

- Additional features
  - Message queues
  - Binary and counting semaphores
  - Mutexes
  - Context switching
    - When kernel switches tasks, it saves context (registers, stack, etc.)
    - When the task is resumed the context is restored

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Recap – FreeRTOS online resources

- freertos.org -> resources
- https://www.freertos.org/Documentation/RTOS_book.html
- Download FreeRTOS for examples!
- More info on memory management https://www.freertos.org/a00111.html

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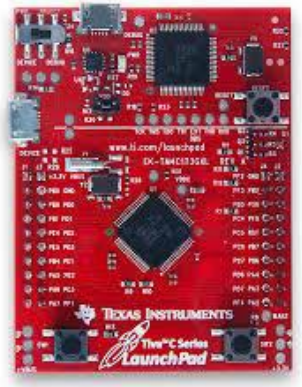
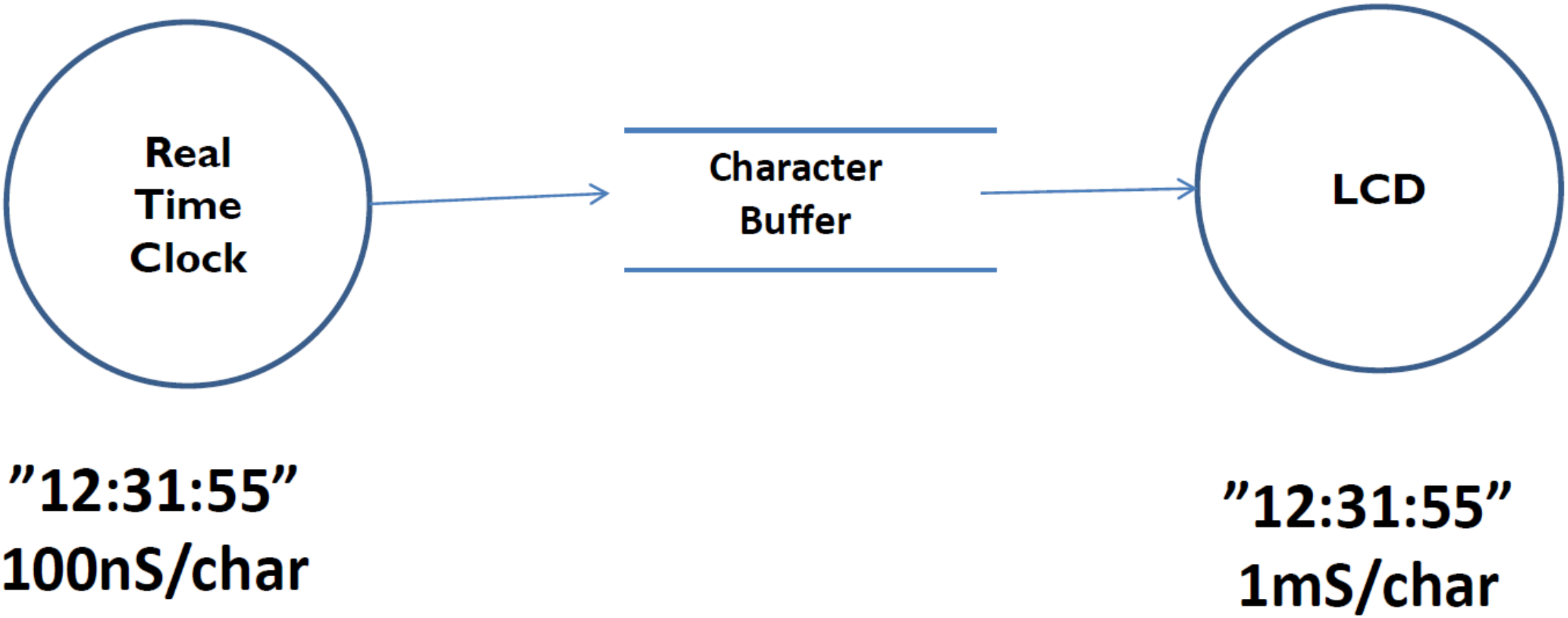
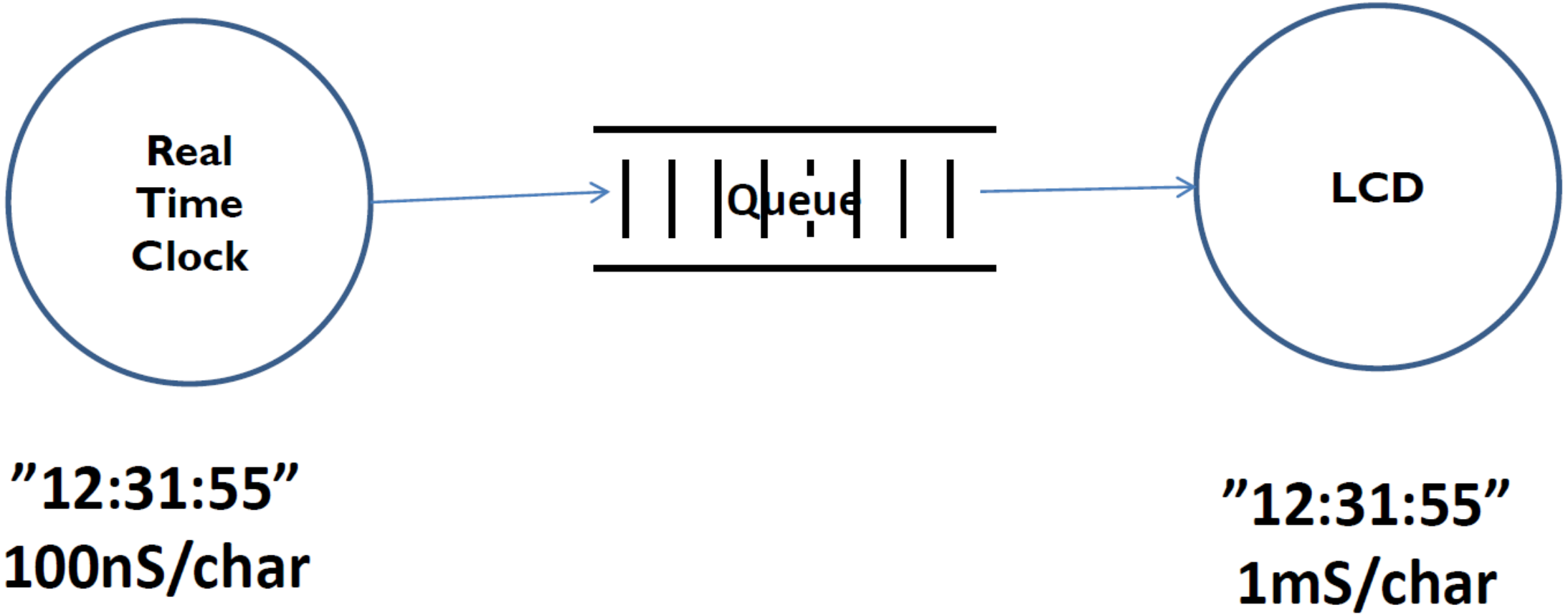
Recap of queues & mutexes
Deadlocks, queues continued



Recap: producer – consumer model



Recap: producer – consumer model



Recap - a queue object

A queue is…
…an individual object in the task diagram
…has a unique name, number, id, key or handle
…implement a FIFO strategy
…offer an API for:

- creating (Create, Open, …)
- insert data (Put, Write, Send, Post, …)
- fetch data (Get, Read, Receive, Retrieve, …)
- show the state of the queue (number of items in the queue) (Full, Empty, Length, …)
- Maybe: read data without removing it from the queue (Peek, …)

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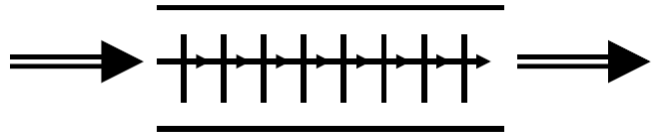
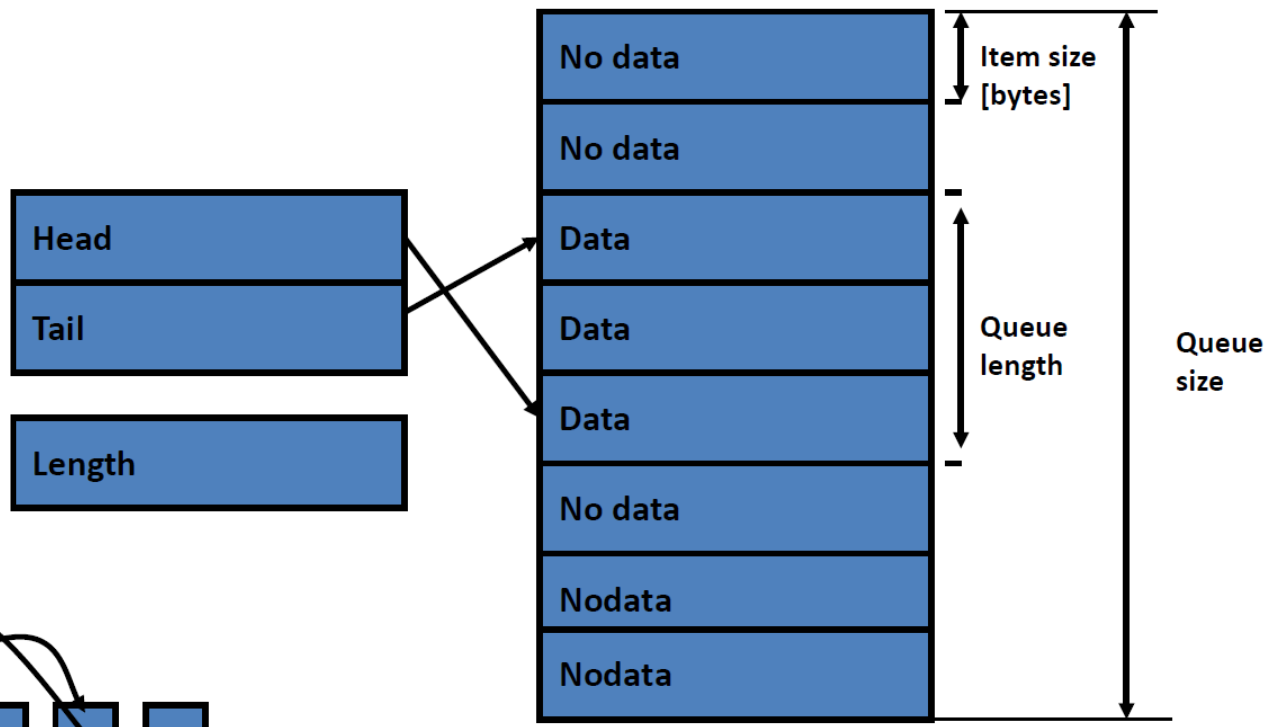
Recap: Queue implementation

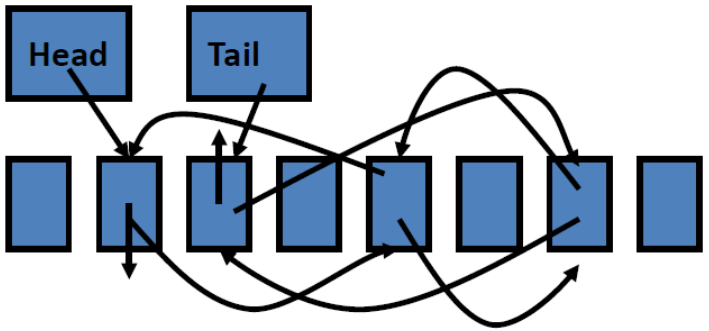
Move data through the queue (like supermarket)



Use pointers (pull a number
-like the post office)



Linked List (like exam
– call the next)



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Recap: queue.c

```
typedef struct
{
            INT8U      head;
            INT8U      tail;
            INT8U      len;
            INT8U buf[QUEUE_SIZE];
} queue_descriptor;
```

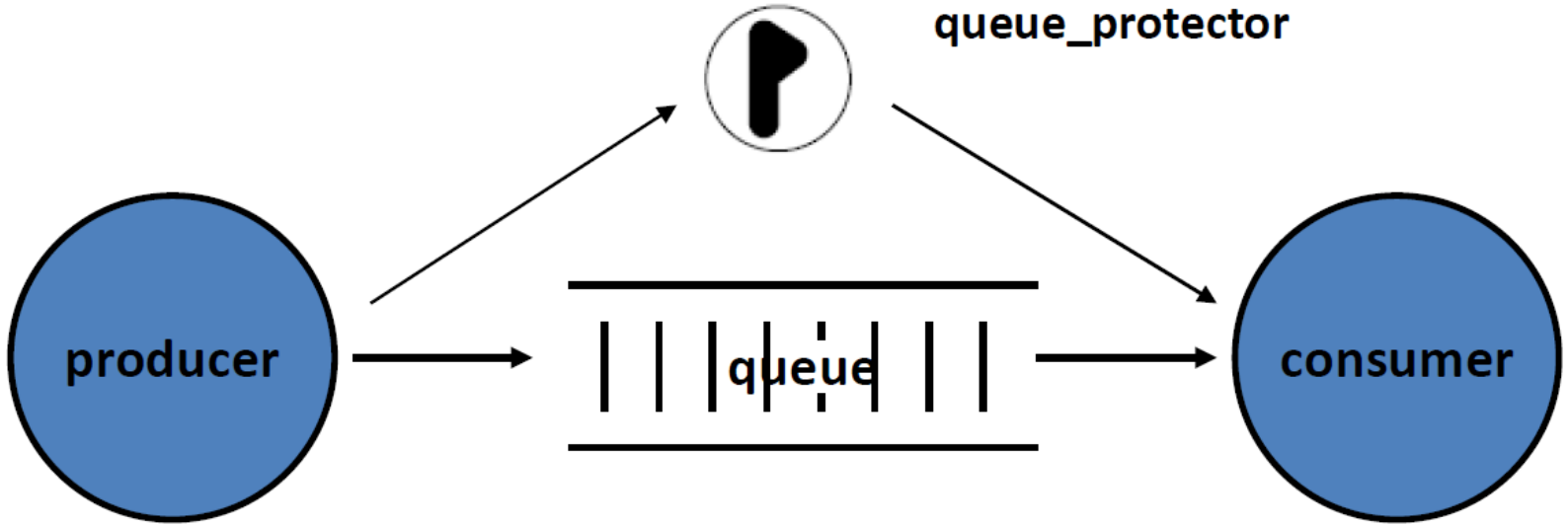
```
queue_descriptor queue_pool[NOF_QUEUES];
```

```
INT8U queue_put( id, ch )
INT8U id;
INT8U ch;
/************************************************
*   Function : See module specification (.h-file).
************************************************/
{
  INT8U result = FALSE;

  if( id < NOF_QUEUES )
  {
    if( queue_pool[id].len < QUEUE_SIZE )
    {
      queue_pool[id].buf[queue_pool[id].tail++] = ch;
      if( queue_pool[id].tail >= QUEUE_SIZE )
        queue_pool[id].tail = 0;
      queue_pool[id].len++;
      result = TRUE;
    }
  }
  return( result );
}
```

```
INT8U queue_get( id )
INT8U id;
/************************************************
*   Function : See module specification (.h-file).
************************************************/
{
  INT8U Ch = '\0';

  if( id < NOF_QUEUES )
  {
    if( queue_pool[id].len )
    {
      Ch = queue_pool[id].buf[queue_pool[id].head++];
      if( queue_pool[id].head >= QUEUE_SIZE )
        queue_pool[id].head = 0;
      queue_pool[id].len--;
    }
  }
  return( Ch );
}
```

Protection of queues

A queue is a shared resource, and as such, it must be protected by a mutex semaphore.



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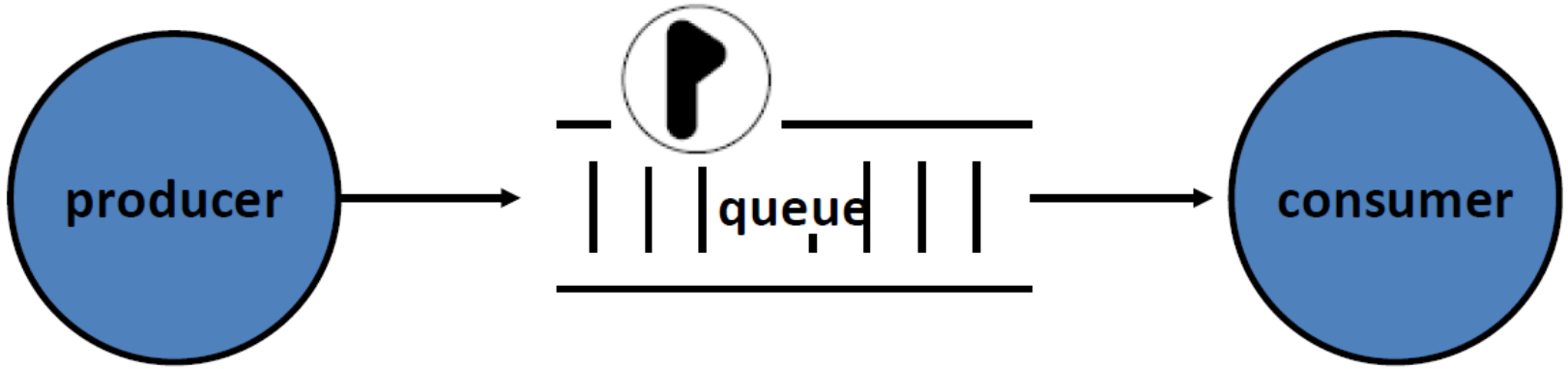
Protection of queues

- Because of that, the mutex is often build into the queue or closely connected to the queue.
- As we are now using pre-emptive scheduling protection of queues is crucial



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Recap: Semaphore - mutex

- Secures mutually exclusive blocks to prevent concurrent access to the same resource
- Binary semaphore
- Creates critical sections

Task A – clock update

```
mutex.wait()
        #critical section
        update(clock)
mutex.signal()
```

Task B – clock display

```
mutex.wait()
        #critical section
        display(clock)
mutex.signal()
```

- Example – contentious meetings with duckie

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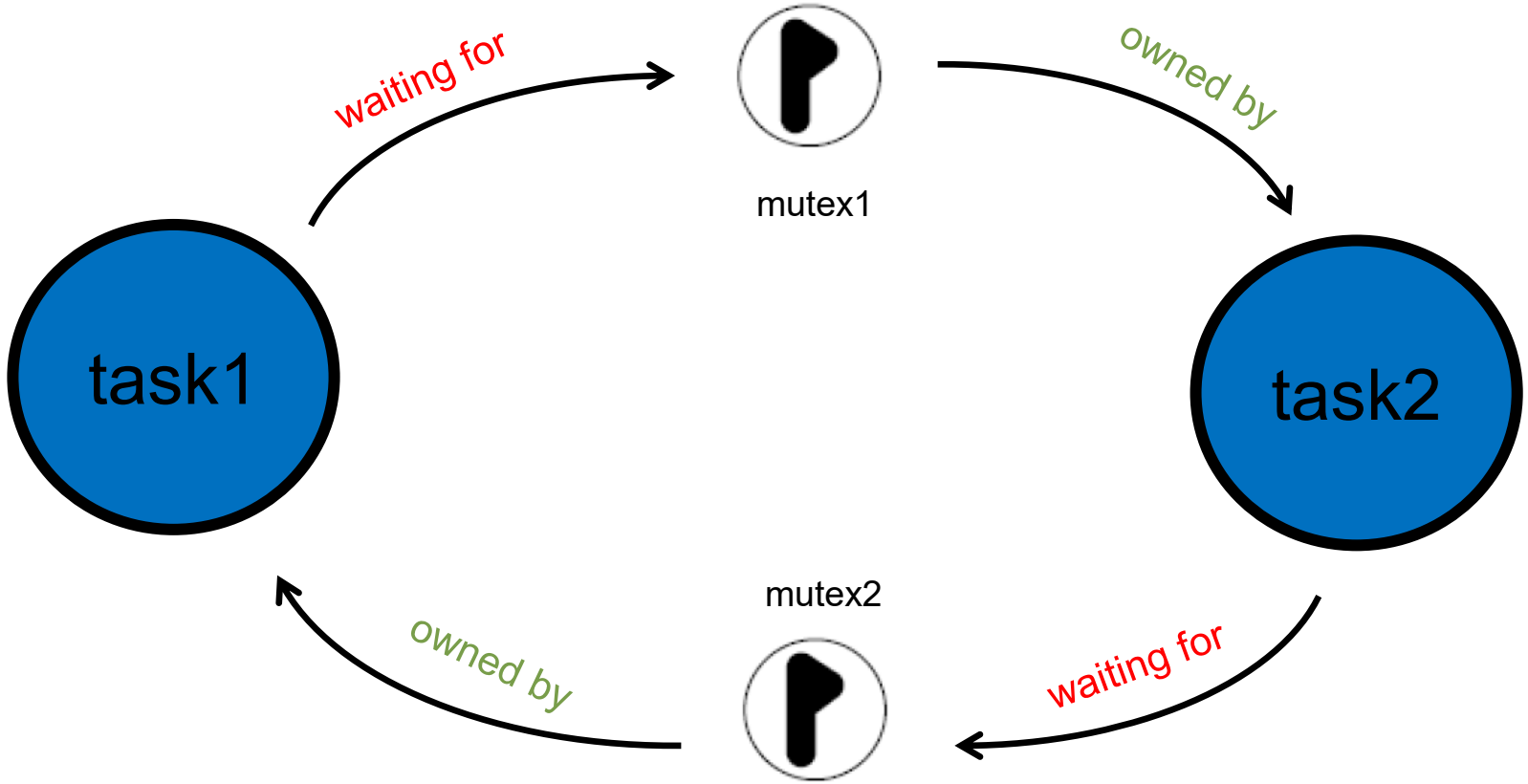
Beware of deadlocks

- Deadlocks appear when two or more tasks are waiting at each other to continue their execution
- This usually happens through resources being blocked with mutexes



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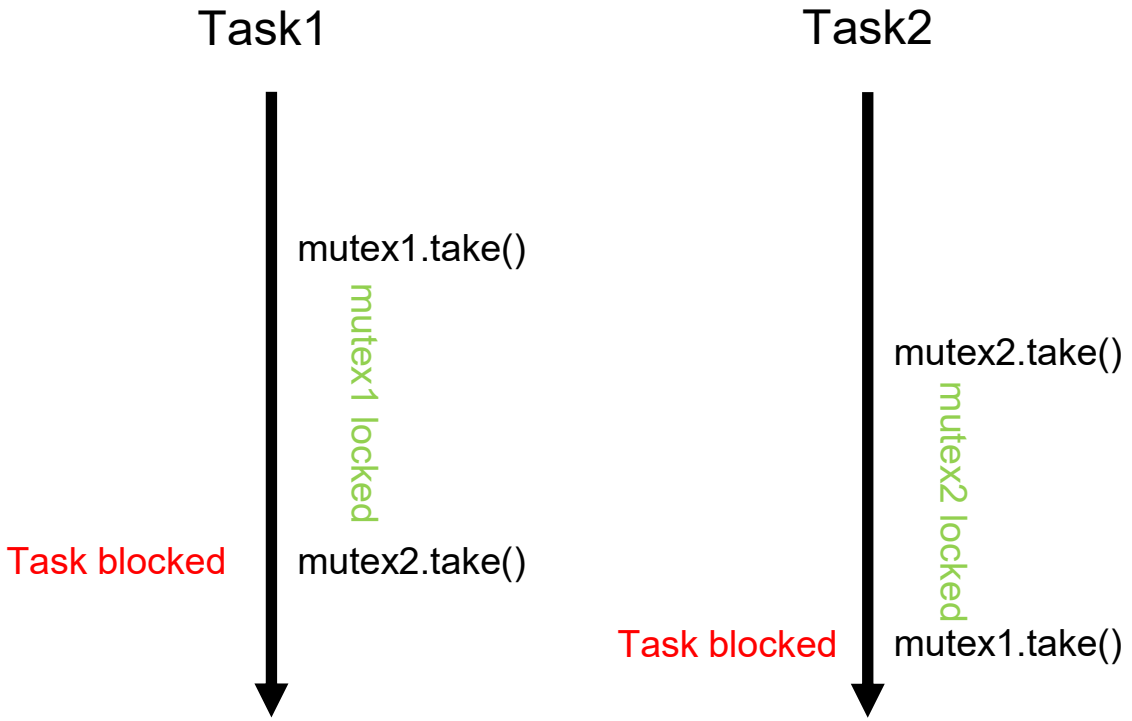
Deadlock in real-life

Deadlock in other words

- A process cannot proceed because it needs to obtain a resource held by another process but it itself is holding a resource that the other process needs



Coffman’s 4 conditions of deadlock

- **Mutual exclusion** – a resource can be held by at most one process
- **Hold and wait -** processes that already hold a resource can wait for another resource
- **Non-preemption** – a resource, once granted, cannot be taken away
- **Circular wait** – two or more processes are waiting for resources held by one of the other processes

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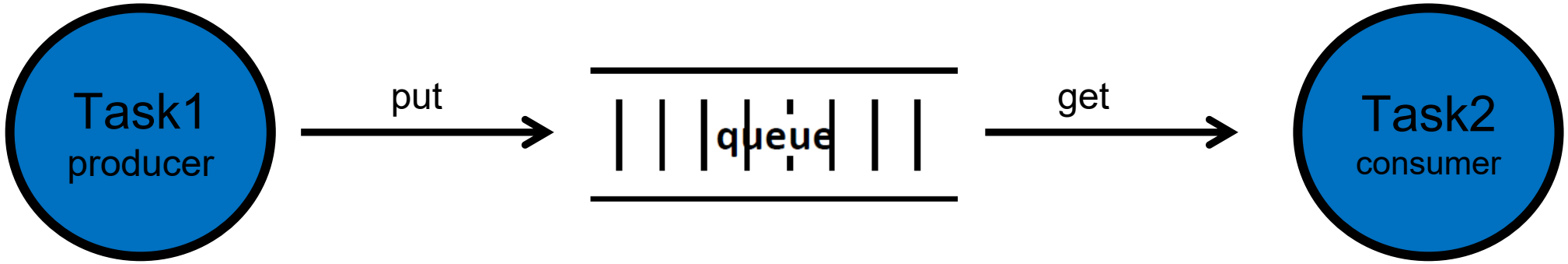
Blocking / non-blocking



- Producer – put()
- If the queue is already full:
  - Throw away the data
  - Do something else and try later
  - Overwrite old data
  - Wait (block)

- Consumer get()
- If the queue is already empty:
  - Do something else and try later
  - Wait (block)

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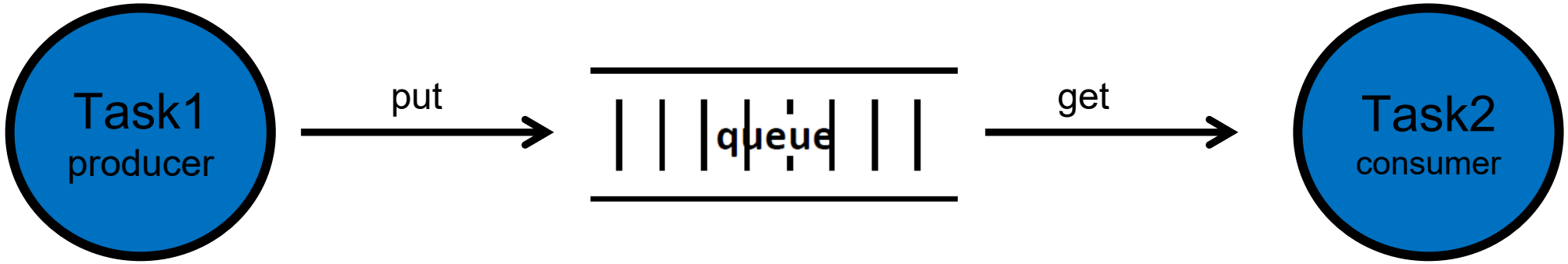
Blocking



- Producer – put()
- Blocking:
  - Forever
  - Timeout

- Consumer get()
- Blocking:
  - Forever
  - Timeout

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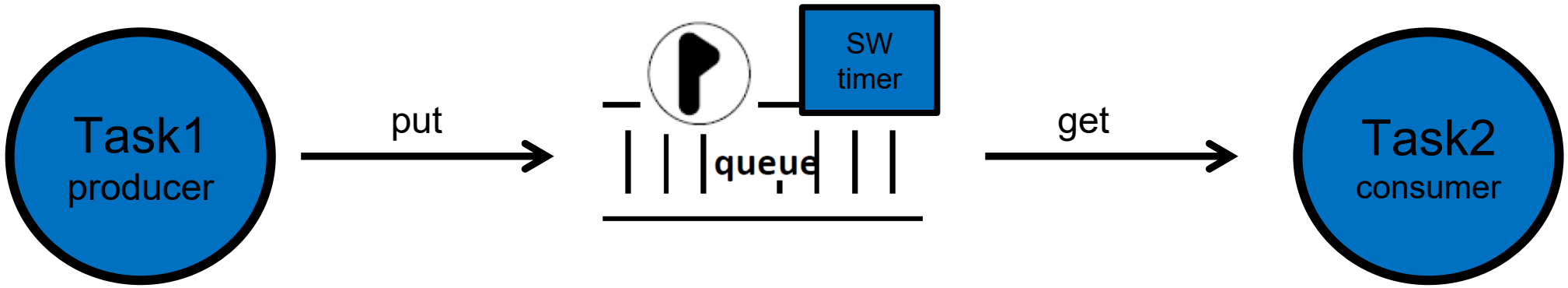
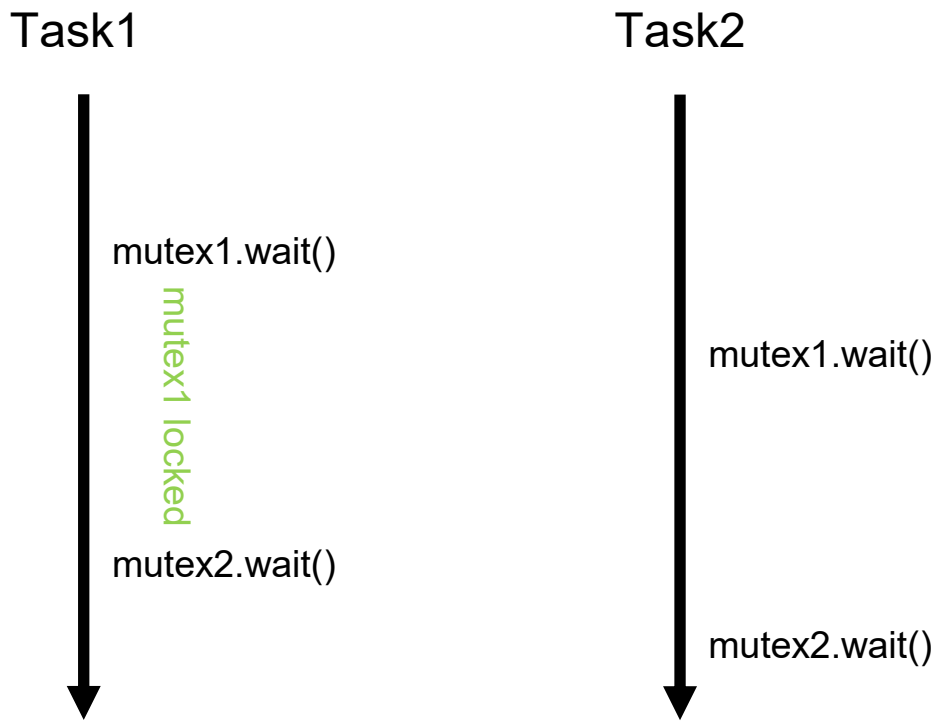
How to resolve? Timeout



How to resolve? Ordering



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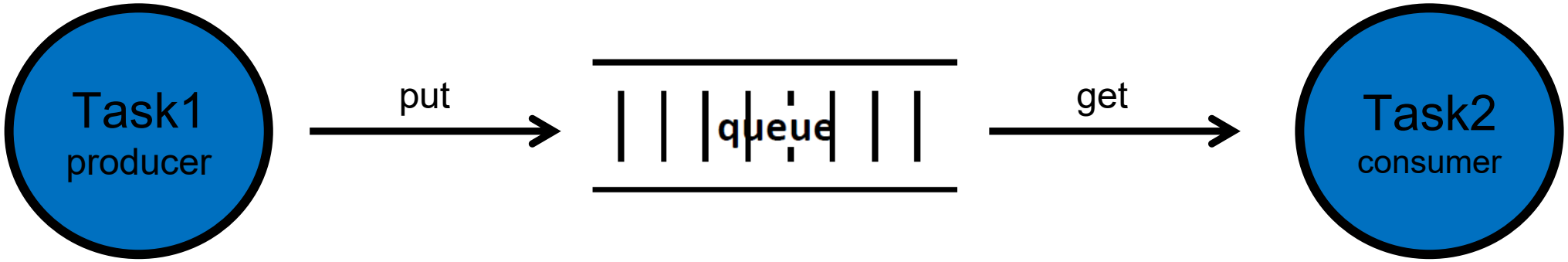
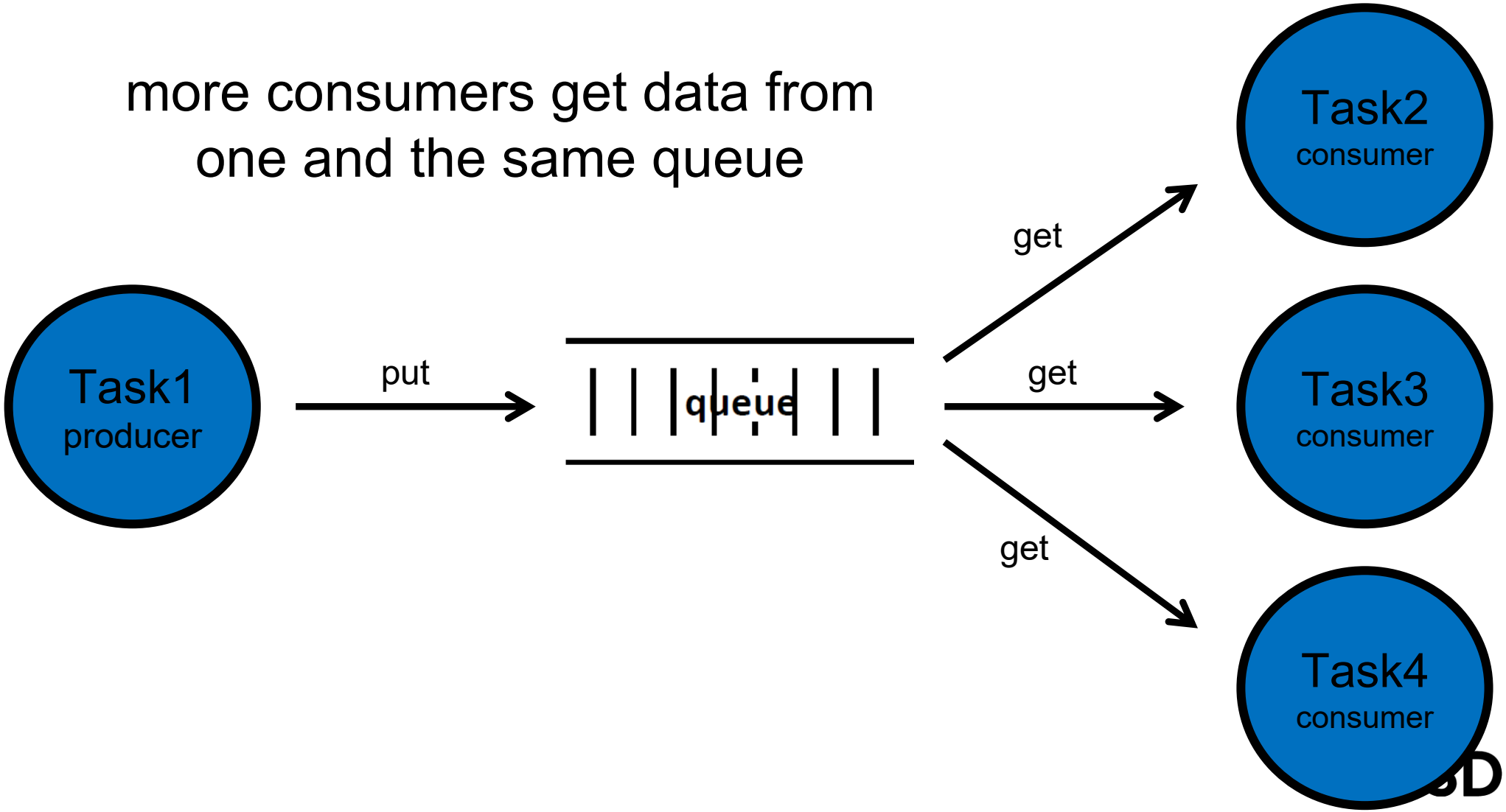
Queue topology

one producer – one consumer



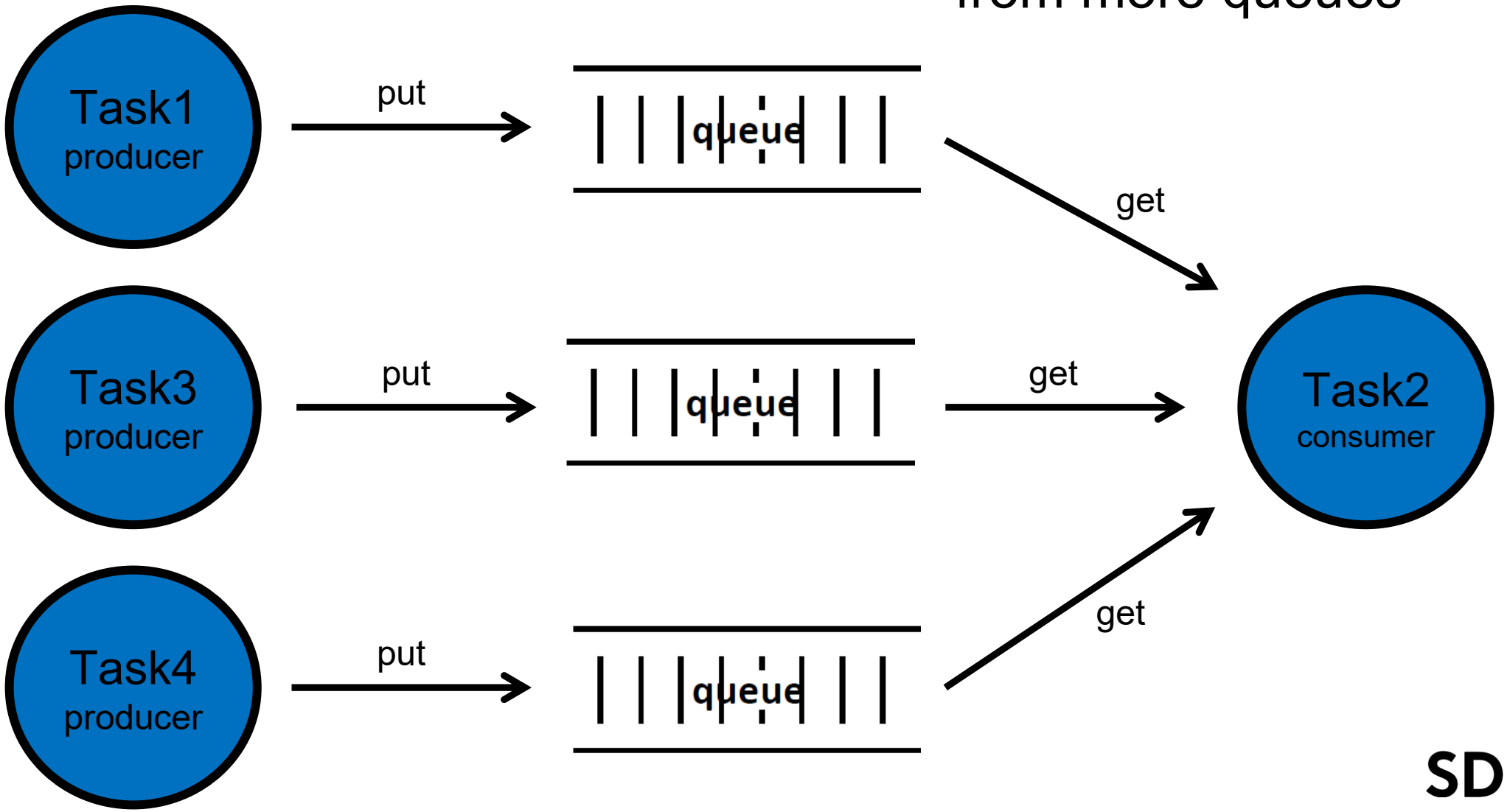
Queue topology

more consumers get data from one and the same queue



Queue topology

One consumer gets data from more queues



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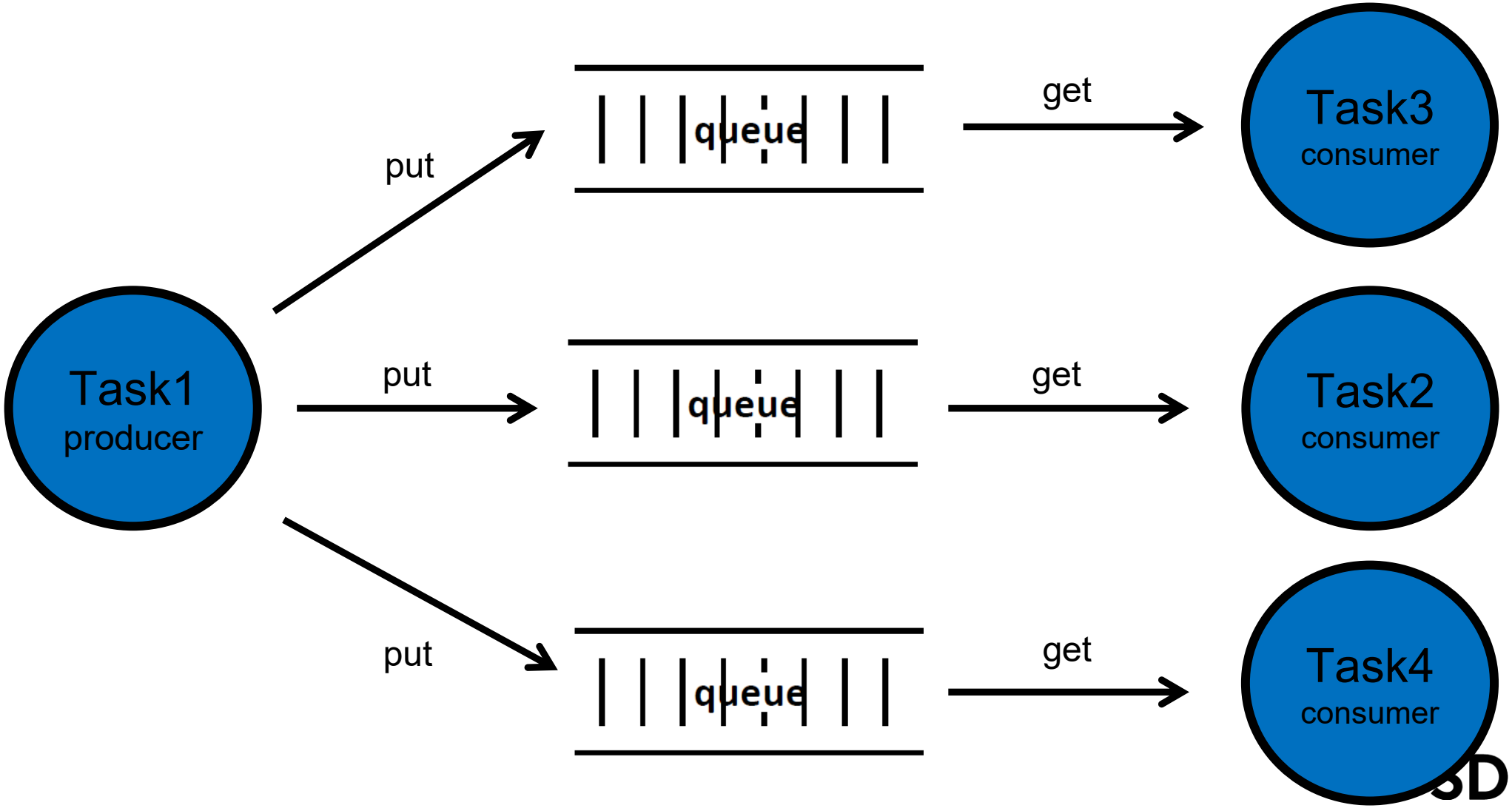
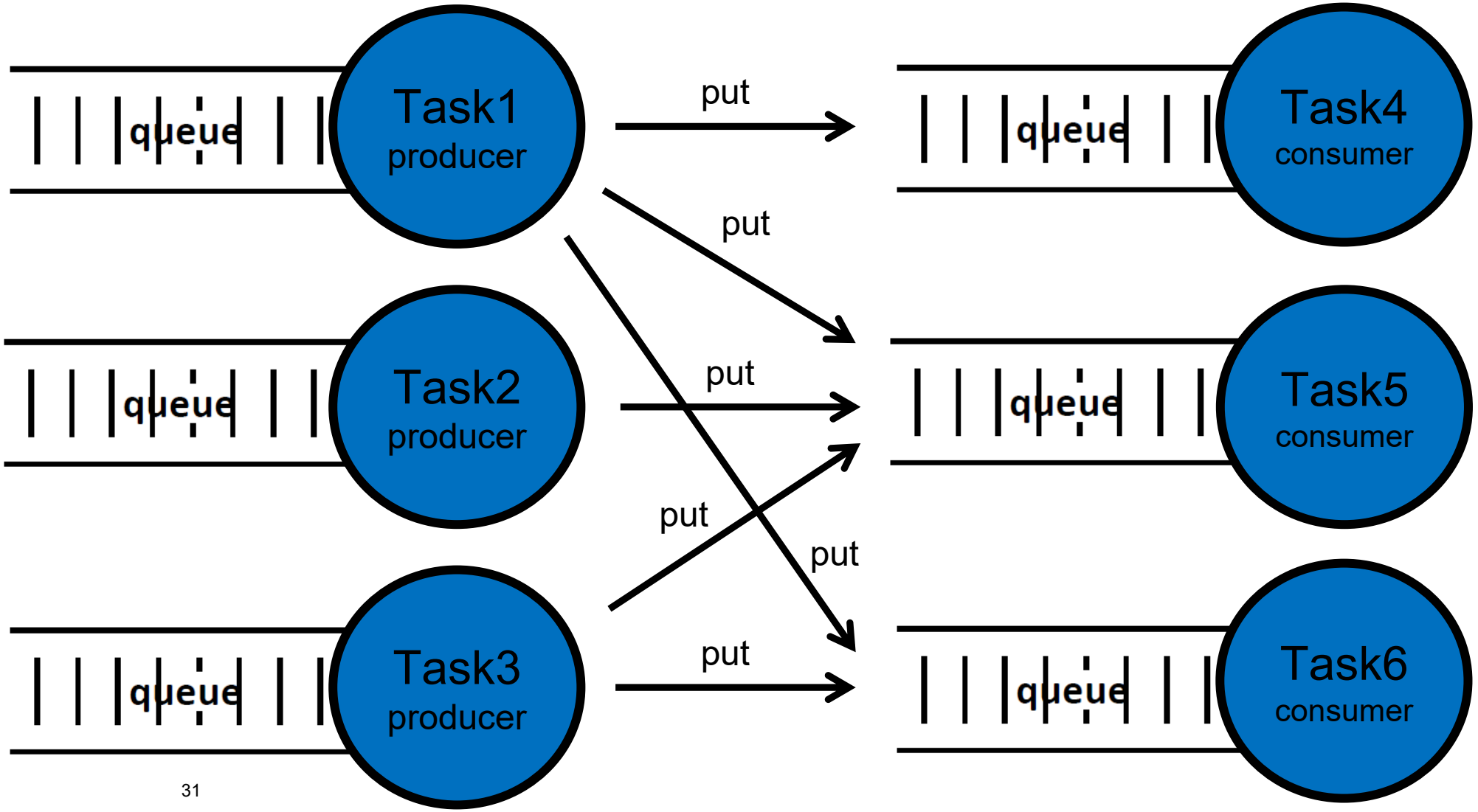
Queue topology

One producer sends data to more queues



Queue topology

Some operating systems implement an input queue to each task



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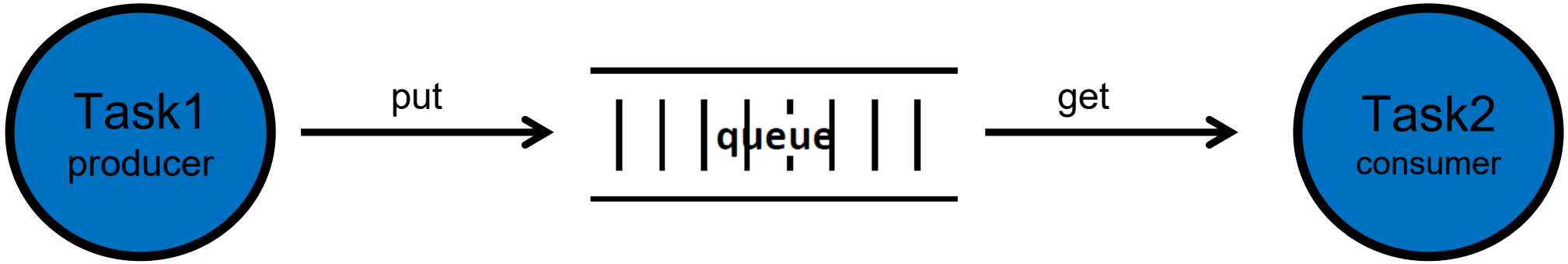
Queues in FreeRTOS



- Queue API
  - xQueueCreate();
  - xQueueSendToBack();
  - xQueueReceive();
  - uxQueueMessagesWaiting();
  - xQueuePeek();
- Queue ID
  - xQueueHandle

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Exercise

- Can you write deadlocking tasks in RTCS? Why?
- Can 2 or 3 tasks that deadlock each other in FreeRTOS?
- How do you resolve such a deadlock?

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Reentrancy

Reentrance

- A function is reentrant or pure if, while it is being executed, can be called again by itself or by another function
- …and it works every time
- i.e. more than one instance of a function can run at the same time

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Errors that occur due to missing reentrance are hard to debug because…

- They seldom show up
- They can provide different faults
- They are dependent on memory configurations

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A function is reentrant if…

- All variables are used atomic or are allocated to the specific instance of the function (not static)
- Common resources like hardware and global variables are used atomic or are protected by task synchronization elements (semaphores)
- It never changes its own code
- It never calls functions which are not reentrant

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Atomic operations

- Ones that cannot be interrupted

Atomic operations

- Statements in C are not necessarily atomic but are typically executed as several assembler instructions

```
Foo++; // becomes
mov ax,[foo]
inc ax
mov [foo],ax;
```

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Atomic operations

- Turn pieces of code atomic by wrapping it in ”interrupt disable”
- disabling interrupts reduces the real-time performance of the system (interrupt latency)
- use instead the known task synchronization methods (e.g. semaphores)

Non reentrant code

```
int foo;
void some_function( void )
{
     foo++;
}
```

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Reentrant code

```
void some_function( void )
{
      int foo;
      foo++;
}
```

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Static variables a problem for reentrance

```
void some_function( void )
{
     static int foo=1;
     foo++;
}
```

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Common resources

- hardware and global variables must be protected by the known task synchronization methods:
  - Semaphores
  - Monitors (abstract data types which allows only one process to execute in a critical section at a time)

Common resources

```
void ChangeIP( unsigned long IP )
{
     Enter( semaphoreIP );
     IP[0] = Byte1(IP);
     IP[1] = Byte2(IP);
     IP[2] = Byte3(IP);
     IP[3] = Byte4(IP);
     Signal( semaphoreIP );
}
```

Avoid calling non-reentrant functions

- Check “old” code
- Beware of 3rd party libraries

Recursion

- a function is recursive, if it calls itself
- a recursive function must necessarily be reentrant
- A reentrant function must not necessarily be recursive

```
double power( double x, int exp )
{
     if( exp <= 0 )
          return(1);
     return( x*power(x, exp-1));
}
```

Writing reentrant code

- Easy to write
- Hard to retrofit (make non-reentrant code reentrant)

Read more about reentrancy

- http://www.ganssle.com/articles/areentra.htm

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